

Chapter 3

Questions: 2.

Problems: 3A.2, 3B.1, and 3B.4

1- Using the continuity equation $\frac{\partial \rho}{\partial t} = -\text{div } \rho \langle v |$, prove:

$$\frac{\partial}{\partial t} \rho \frac{v^2}{2} + \text{div } \rho \frac{v^2}{2} \langle v | = \rho \frac{D}{Dt} \frac{v^2}{2}$$

$$\frac{\partial}{\partial t} \rho |v\rangle + | \text{div } \rho |v\rangle \langle v | \rangle = \rho \frac{D}{Dt} |v\rangle$$

2- Using cartesian coordinates, prove for a fluid of constant density

$$\text{div } |\nabla v|^{\dagger} = 0$$

1- Using the continuity equation $\frac{\partial \rho}{\partial t} = -\text{div } \rho \langle v |$, prove:

$$a) \quad \frac{\partial}{\partial t} \rho \frac{v^2}{2} + \text{div } \rho \frac{v^2}{2} \langle v | = \rho \frac{D}{Dt} \frac{v^2}{2}$$

$$b) \quad \frac{\partial}{\partial t} \rho |v\rangle + | \text{div } \rho |v\rangle \langle v | \rangle = \rho \frac{D}{Dt} |v\rangle$$

$$a) \quad \frac{\partial \rho}{\partial t} = -\text{div } \rho \langle v |$$

By Definition:

$$\frac{D}{Dt} \left(\frac{v^2}{2} \right) = \frac{\partial}{\partial t} \left(\frac{v^2}{2} \right) + \langle v | \cdot \nabla \left(\frac{v^2}{2} \right)$$

$$\Rightarrow \int \frac{D}{Dt} \left(\frac{v^2}{2} \right) = \int \frac{\partial}{\partial t} \left(\frac{v^2}{2} \right) + \int \langle v | \cdot \nabla \left(\frac{v^2}{2} \right)$$

$$\Rightarrow \underbrace{\frac{\partial}{\partial t} \left(\int \frac{v^2}{2} \right)} + \underbrace{\text{div} \left(\int \frac{v^2}{2} \langle v | \right)}$$

$$\frac{\partial}{\partial t} \left(\int \frac{v^2}{2} \right) = \frac{v^2}{2} \frac{\partial \rho}{\partial t} + \rho \frac{\partial}{\partial t} \left(\frac{v^2}{2} \right)$$

$$\text{div} \left(\int \frac{v^2}{2} \langle v | \right) = \frac{v^2}{2} \text{div}(\rho \langle v |) + \rho \langle v | \cdot \nabla \left(\frac{v^2}{2} \right)$$

$$\frac{\partial}{\partial t} \left(\int \frac{v^2}{2} \right) + \text{div} \left(\int \frac{v^2}{2} \langle v | \right) = \frac{v^2}{2} \left[\frac{\partial \rho}{\partial t} + \text{div}(\rho \langle v |) \right] + \rho \left[\frac{\partial}{\partial t} \left(\frac{v^2}{2} \right) + \langle v | \cdot \nabla \left(\frac{v^2}{2} \right) \right]$$

$$\text{we know: } \frac{\partial \rho}{\partial t} + \text{div } \rho \langle v | = 0$$

$$\frac{\partial}{\partial t} \left(\int \frac{v^2}{2} \right) + \text{div} \left(\int \frac{v^2}{2} \langle v | \right) = \int \left[\frac{\partial}{\partial t} \left(\frac{v^2}{2} \right) + \langle v | \cdot \nabla \left(\frac{v^2}{2} \right) \right] = \int \frac{D}{Dt} \left(\frac{v^2}{2} \right)$$

b) Prove $\frac{\partial \rho}{\partial t} \langle v \rangle + \langle \text{div}(\rho \langle v \rangle \langle v \rangle) \rangle = \rho \frac{D}{Dt} \langle v \rangle$

take $\frac{D}{Dt} \langle v \rangle = \frac{\partial \langle v \rangle}{\partial t} + \langle v \rangle \cdot \nabla \langle v \rangle$

$\rho \Rightarrow \rho \frac{D}{Dt} \langle v \rangle = \rho \frac{\partial \langle v \rangle}{\partial t} + \rho \langle v \rangle \cdot \nabla \langle v \rangle$

$\Rightarrow \frac{\partial}{\partial t} (\rho \langle v \rangle) + \langle \text{div}(\rho \langle v \rangle \langle v \rangle) \rangle$

$\Rightarrow \langle v \rangle \frac{\partial \rho}{\partial t} + \rho \frac{\partial \langle v \rangle}{\partial t} + \langle v \rangle \cdot \text{div}(\rho \langle v \rangle) + \rho \langle v \rangle \cdot \nabla \langle v \rangle$

$\frac{\partial}{\partial t} (\rho \langle v \rangle) + \langle \text{div}(\rho \langle v \rangle \langle v \rangle) \rangle = \langle v \rangle \left[\frac{\partial \rho}{\partial t} + \text{div}(\rho \langle v \rangle) \right] + \rho \left[\frac{\partial \langle v \rangle}{\partial t} + \langle v \rangle \cdot \nabla \langle v \rangle \right]$

$\Rightarrow$ we know $\frac{\partial \rho}{\partial t} + \text{div}(\rho \langle v \rangle) = 0$

$\Rightarrow \frac{\partial}{\partial t} (\rho \langle v \rangle) + \langle \text{div}(\rho \langle v \rangle \langle v \rangle) \rangle = \rho \left[\frac{\partial \langle v \rangle}{\partial t} + \langle v \rangle \cdot \nabla \langle v \rangle \right]$

we know $\frac{D \langle v \rangle}{Dt} = \frac{\partial \langle v \rangle}{\partial t} + \langle v \rangle \cdot \nabla \langle v \rangle$

$\Rightarrow \frac{\partial}{\partial t} (\rho \langle v \rangle) + \langle \text{div}(\rho \langle v \rangle \langle v \rangle) \rangle = \rho \frac{D \langle v \rangle}{Dt}$

2- Using cartesian coordinates, prove for a fluid of constant density

$$\operatorname{div} |\nabla v|^{\dagger} = 0$$

$$\operatorname{div} |\nabla v|^{\dagger} = 0$$

$$\operatorname{div} \langle v \rangle = 0 \quad \Rightarrow \quad \left[\frac{\partial v_x}{\partial x} + \frac{\partial v_y}{\partial y} + \frac{\partial v_z}{\partial z} \right] = 0$$

take ∇
of both
sides

$$\nabla(\operatorname{div} \langle v \rangle) = 0$$

$$\frac{\partial^2 v_x}{\partial x_i \partial x} + \frac{\partial^2 v_y}{\partial x_i \partial y} + \frac{\partial^2 v_z}{\partial x_i \partial z} = 0$$

$$\Rightarrow \operatorname{div} (|\nabla v|^{\dagger}) = 0$$

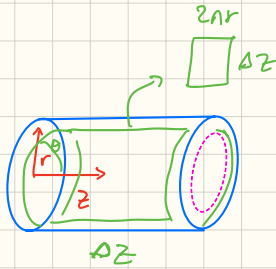
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2. By making a mass balance over a volume element $(\Delta r)(r\Delta\theta)(\Delta z)$ derive the equation of continuity in cylindrical coordinates.

div in cylindrical coordinates:

$$\text{div } \rho \mathbf{v}: \frac{1}{r} \frac{\partial}{\partial r}(r \rho v_r) + \frac{1}{r} \frac{\partial}{\partial \theta}(\rho v_\theta) + \frac{\partial}{\partial z}(\rho v_z)$$



$$V = \Delta r(r\Delta\theta)\Delta z$$

proof: $\text{div } \rho \mathbf{v} = \lim_{V \rightarrow 0} \frac{1}{V} \int_S ds \rho \mathbf{v} \cdot \mathbf{n}$

$$\begin{aligned} \int_S ds \rho \mathbf{v} \cdot \mathbf{n} = & - [\rho v_r r \Delta\theta \Delta z]_r + [\rho v_r r \Delta\theta \Delta z]_{r+\Delta r} - [\rho v_\theta r \Delta r \Delta\theta \Delta z]_\theta + [\rho v_\theta r \Delta r \Delta\theta \Delta z]_{\theta+\Delta\theta} \\ & - [\rho v_z r \Delta r \Delta\theta \Delta z]_z + [\rho v_z r \Delta r \Delta\theta \Delta z]_{z+\Delta z} \end{aligned}$$

Divide by volume: $r \Delta r \Delta\theta \Delta z$
 $\Rightarrow$ and take $\lim_{V \rightarrow 0}$

$$\begin{aligned} \Rightarrow \int_S ds \rho \mathbf{v} \cdot \mathbf{n} = & - \frac{[\rho v_r r \Delta\theta \Delta z]_r}{r \Delta r \Delta\theta \Delta z} + \frac{[\rho v_r r \Delta\theta \Delta z]_{r+\Delta r}}{r \Delta r \Delta\theta \Delta z} - \frac{[\rho v_\theta r \Delta r \Delta\theta \Delta z]_\theta}{r \Delta r \Delta\theta \Delta z} + \frac{[\rho v_\theta r \Delta r \Delta\theta \Delta z]_{\theta+\Delta\theta}}{r \Delta r \Delta\theta \Delta z} \\ & - \frac{[\rho v_z r \Delta r \Delta\theta \Delta z]_z}{r \Delta r \Delta\theta \Delta z} + \frac{[\rho v_z r \Delta r \Delta\theta \Delta z]_{z+\Delta z}}{r \Delta r \Delta\theta \Delta z} \end{aligned}$$

$$\Rightarrow \int_S ds \rho \mathbf{v} \cdot \mathbf{n} = \frac{1}{r} \frac{\partial}{\partial r}(r \rho v_r) + \frac{1}{r} \frac{\partial}{\partial \theta}(\rho v_\theta) + \frac{\partial}{\partial z}(\rho v_z)$$

We know: $-\text{div } \rho \mathbf{v} = \frac{\partial \rho}{\partial t}$

$$\Rightarrow \frac{\partial \rho}{\partial t} + \left[\frac{1}{r} \frac{\partial}{\partial r}(r \rho v_r) + \frac{1}{r} \frac{\partial}{\partial \theta}(\rho v_\theta) + \frac{\partial}{\partial z}(\rho v_z) \right] = 0$$

Continuity Equation /
Mass balance in cylindrical
coordinates

3A.2 Friction loss in bearings.¹ Each of two screws on a large motor-ship is driven by a 4000-hp engine. The shaft that connects the motor and the screw is 16 in. in diameter

and rests in a series of sleeve bearings that give a 0.005 in. clearance. The shaft rotates at 50 rpm, the lubricant has a viscosity of 5000 cp, and there are 20 bearings, each 1 ft in length. Estimate the fraction of engine power expended in rotating the shafts in their bearings. Neglect the effect of the eccentricity.

Answer: 0.115

from 3A.1: $P = \text{torque} \cdot \omega_i = 4\pi\mu\omega_i^2 R^2 L \left(\frac{k^2}{1-k^2} \right)$: power loaded on bearing

2 shafts per bearing
20 bearings for each shaft } $\Rightarrow L = 2 \times 20 = 40 \text{ ft.}$

Convert units:

$$\omega_i = 50 \text{ rpm} \left| \frac{1 \text{ min}}{60 \text{ s}} \right| \left| 2\pi \frac{\text{rad}}{\text{min}} \right| = \frac{5\pi}{3} \text{ rad/s}$$

$$L = 40 \text{ ft} = 12.19 \text{ m}$$

$$\mu = 5000 \text{ cp} = 5 \text{ kg/ms}$$

$$k = \frac{16}{16 + 2(0.005)} = 0.999775$$

$$R = \frac{B}{39.37 \text{ in}} = 0.2032 \text{ m (radius)}$$

Solve for Power: $P = 4\pi\mu\omega_i^2 R^2 L \left(\frac{k^2}{1-k^2} \right)$

$$P = 4\pi(5.0 \frac{\text{kg}}{\text{ms}}) \left(\frac{5\pi}{3} \frac{\text{rad}}{\text{s}} \right)^2 (0.2032 \text{ m})^2 (12.19 \text{ m}) \left(\frac{0.999775^2}{1 - 0.999775^2} \right)$$

$$P = 6.93 \cdot 10^5 \frac{\text{kg m}^2}{\text{s}^3} = 6.93 \cdot 10^5 \frac{\text{J}}{\text{s}} = 6.93 \cdot 10^2 \text{ kW}$$

$$\text{Power} = 6.93 \cdot 10^5 \text{ W} \left| \frac{1 \text{ hp}}{745.7 \text{ kW}} \right| = 929.33 \text{ hp}$$

Fraction of available power lost in bearing friction:

$$\frac{929.33}{(4000 + 929.33)} = 0.116$$

3B.1 Flow between coaxial cylinders and concentric spheres.

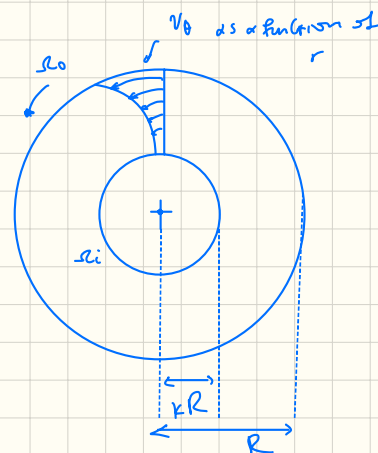
(a) The space between two coaxial cylinders is filled with an incompressible fluid at constant temperature. The radii of the inner and outer wetted surfaces are κR and R , respectively. The angular velocities of rotation of the inner and outer cylinders are Ω_i and Ω_o . Determine the velocity distribution in the fluid and the torques on the two cylinders needed to maintain the motion.

(b) Repeat part (a) for two concentric spheres.

Answers:

$$(a) v_\theta = \frac{\kappa R}{1 - \kappa^2} \left[(\Omega_o - \Omega_i \kappa^2) \left(\frac{r}{\kappa R} \right) + (\Omega_i - \Omega_o) \left(\frac{\kappa R}{r} \right) \right]$$

$$(b) v_\phi = \frac{\kappa R}{1 - \kappa^3} \left[(\Omega_o - \Omega_i \kappa^3) \left(\frac{r}{\kappa R} \right) + (\Omega_i - \Omega_o) \left(\frac{\kappa R}{r} \right)^2 \right] \sin \theta$$



a) Preliminary work:

$$v_r = 0 = v_z$$

$$v_\theta = v_\theta(r, \theta, z) \quad \text{S.S.}$$

$$p = p(r, z)$$

Navier Stokes Eqn (fun of motion constant ρ, μ):

θ -comp:

$$\rho \left(\frac{\partial v_\theta}{\partial t} + v_r \frac{\partial v_\theta}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} + v_z \frac{\partial v_\theta}{\partial z} + \frac{v_\theta}{r} \right) = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{\partial}{\partial r} \left(\frac{1}{r} \frac{\partial}{\partial r} (r v_\theta) \right) + \frac{1}{r^2} \frac{\partial^2 v_\theta}{\partial \theta^2} + \frac{\partial^2 v_\theta}{\partial z^2} + \frac{2}{r^2} \frac{\partial v_r}{\partial \theta} \right] + \rho g_\theta$$

$$\Rightarrow \int 0 = \int \frac{d}{dr} \left(\frac{1}{r} \frac{d}{dr} (r v_\theta) \right)$$

$$\Rightarrow \frac{1}{r} \frac{d}{dr} (r v_\theta) = C_1$$

$$\Rightarrow \int \frac{d}{dr} (r v_\theta) = \int C_1 \cdot r$$

$$\Rightarrow r v_\theta = \frac{C_1 r^2}{2} + C_2$$

$$\Rightarrow v_\theta = \frac{C_1}{2} r + \frac{C_2}{r}$$

$$, \text{ let } C_1 = \frac{C_1}{2} \Rightarrow$$

$$v_\theta = C_1' \cdot r + \frac{C_2}{r}$$

BCs: $\text{at } r = \kappa R \Rightarrow v_\theta = \Omega_i \cdot \kappa R$
 $\text{at } r = R \Rightarrow v_\theta = \Omega_o R$

$$v_\theta(\kappa R) = c'_1 \cdot \kappa R + \frac{c_2}{\kappa R}$$

$$\Omega_i \kappa R = c'_1 \kappa R + \frac{c_2}{\kappa R}$$

$$\Omega_i = c'_1 + \frac{c_2}{(\kappa R)^2}$$

$$v_\theta(R) = c'_1 \cdot R + \frac{c_2}{R}$$

$$\Omega_o R = c'_1 R + \frac{c_2}{R}$$

$$\Omega_o = c'_1 + \frac{c_2}{R^2}$$

$$\Rightarrow \Omega_i - \Omega_o = \frac{c_2}{R^2} \left(\frac{1}{\kappa^2} - 1 \right) = \frac{c_2}{R^2} \left(\frac{1 - \kappa^2}{\kappa^2} \right)$$

$$\Rightarrow c_2 = (\Omega_i - \Omega_o) \frac{(\kappa R)^2}{1 - \kappa^2}, \quad c'_1 = \frac{\Omega_o - \Omega_i \kappa^2}{1 - \kappa^2}$$

$\Rightarrow$ VELOCITY DISTRIBUTION:

$$\frac{v_\theta}{r} = \frac{(\Omega_o - \Omega_i \kappa^2)}{1 - \kappa^2} - (\Omega_o - \Omega_i) \left(\frac{\kappa^2}{1 - \kappa^2} \right) \left(\frac{R}{r} \right)^2$$

To find torque: $\tau \cdot A \cdot \text{lever arm}$

$$\tau_{r\theta} = -\mu r \frac{d}{dr} \left(\frac{v_\theta}{r} \right) = \mu r (\Omega_o - \Omega_i) \frac{(\kappa R)^2}{1 - \kappa^2} \left(\frac{2}{r^3} \right)$$

$$\tau_{r\theta} = -2\mu (\Omega_o - \Omega_i) \left(\frac{\kappa^2}{1 - \kappa^2} \right) \left(\frac{R}{r} \right)^2$$

Torque on outer former cylinders: (z-axis)

$$\tau_{z, \text{outer}} = \int_0^L \int_0^{2\pi} (-\tau_{r\theta}) R \Big|_{r=R} \cdot R d\theta dz = (-\tau_{r\theta}) \Big|_{r=R} (2\pi R L) \cdot R$$

$$= 2\pi R^2 L \left[2\mu (\Omega_o - \Omega_i) \left(\frac{\kappa^2}{1 - \kappa^2} \right) \left(\frac{R}{r} \right)^2 \right]_{r=R}$$

$$= 4\pi \mu L (\Omega_o - \Omega_i) \frac{(\kappa R)^2}{1 - \kappa^2}$$

$$\begin{aligned}
 \tau_{z, \text{inner}} &: \int_0^L \int_0^{2\pi} (\tau_{r\theta} k r) \Big|_{r=kR} \cdot k r d\theta dz \\
 &= +\tau_{r\theta} \Big|_{r=kR} (2\pi k r L) (k r) \\
 &= 2\pi k^2 R^2 L \left[-2\mu (r_0 - r_i) \left(\frac{k^2}{1-k^2} \right) \left(\frac{R}{r} \right)^2 \right]_{kR} \\
 &= 4\pi\mu L \left[(r_i - r_0) \frac{(kR)^2}{1-k^2} \right]
 \end{aligned}$$

VELOCITY DISTRIBUTION:

$$v_\theta = \left(\frac{kR}{1-k^2} \right) \left[(r_0 - r_i k^2) \left(\frac{r}{kR} \right) + (r_i - r_0) \left(\frac{kR}{r} \right) \right]$$

b) see textbook figure 3.6-10

Postulates:

- 1) S.S
- 2) $v_r = v_\theta = 0$
- 3) $v_\phi = v_\phi(r, \theta)$
- 4) $g_\phi = 0$
- 5) $\beta = \beta(r, \theta)$

Navier Stokes eqn ϕ comp:

$$\begin{aligned}
 \rho \left(\frac{\partial v_\phi}{\partial t} + v_r \frac{\partial v_\phi}{\partial r} + \frac{v_\theta}{r} \frac{\partial v_\phi}{\partial \theta} + \frac{v_\phi}{r \sin \theta} \frac{\partial v_\phi}{\partial \phi} + \frac{v_\phi v_\theta}{r} + \frac{v_\phi v_r \cot \theta}{r} \right) &= -\frac{1}{r \sin \theta} \frac{\partial p}{\partial \phi} \\
 + \mu \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (v_\phi \sin \theta) \right) + \frac{1}{r^2 \sin^2 \theta} \frac{\partial^2 v_\phi}{\partial \phi^2} + \frac{2}{r^2 \sin \theta} \frac{\partial v_r}{\partial \phi} + \frac{2 \cot \theta}{r^2 \sin \theta} \frac{\partial v_\theta}{\partial \phi} \right] &= \rho g_\phi \quad (\text{B.6-9})
 \end{aligned}$$

$$\Rightarrow 0 = \mu \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\phi}{\partial r} \right) + \frac{1}{r^2} \frac{\partial}{\partial \theta} \left(\frac{1}{\sin \theta} \frac{\partial}{\partial \theta} (v_\phi \sin \theta) \right) \right]$$

$$\Rightarrow \left[\frac{d}{dr} \left(r^2 \frac{df}{dr} \right) - f = 0 \Rightarrow f = c_1 r + \frac{c_2}{r^2} \right]$$

$$\Rightarrow v_\phi(r, \theta) = c_1' r + \frac{c_2}{r^3} \sin \theta \Rightarrow \frac{v_\phi}{r \sin \theta} = \frac{c_1'}{\sin \theta} + \frac{c_2}{r^3}$$

$$\begin{aligned}
 \text{B.C: } r=kR \quad v_\phi(r, \theta) &= r_i k R \cdot \sin \theta \\
 r=R \quad v_\phi(r, \theta) &= r_0 R \sin \theta
 \end{aligned}$$

$$\Rightarrow \frac{r_i}{\sin \theta} = \frac{c_1'}{(kR)^3} + \frac{c_2}{(kR)^3} \quad \text{and} \quad \frac{r_0}{\sin \theta} = \frac{c_1'}{R^3} + \frac{c_2}{R^3}$$

$$\Rightarrow C_2 = (\Omega_i - \Omega_o) \frac{(kR)^3}{1-k^3}, \quad C_1' = \frac{(\Omega_o - \Omega_i k^3)}{1-k^3} \sin \theta$$

shear stress:

$$\tau_{r\phi} = -\mu r \frac{d}{dr} \left(\frac{v_\phi}{r} \right) = -\mu r (\Omega_o - \Omega_i) \frac{(kR)^3}{1-k^3} \left(\frac{3}{r^4} \right)$$

$$\tau_{r\phi} = -3\mu (\Omega_o - \Omega_i) \left(\frac{k^3}{1-k^3} \right) \left(\frac{R}{r} \right)^3$$

Torque outer wall:

$$T_{\text{outer}} = \int_0^{2\pi} \int_0^\pi (-\tau_{r\phi} R \sin \theta) \Big|_{r=R} \cdot R^2 \sin \theta \, d\theta \, d\phi$$

$$= \int_0^{2\pi} \int_0^\pi \left[3\mu (\Omega_o - \Omega_i) \left(\frac{k^3}{1-k^3} \right) \left(\frac{R}{r} \right)^3 \sin \theta \right]_{r=R} R^3 \sin \theta \, d\theta \, d\phi$$

$$\star \int_0^\pi \sin^2 \theta \, d\theta = \frac{\pi}{2}$$

$$\int_0^{2\pi} d\phi = 2\pi$$

$$= 3\mu (\Omega_o - \Omega_i) \frac{(kR)^3}{1-k^3} \left(\frac{\pi}{2} \right) 2\pi$$

$$= 8\pi\mu (\Omega_o - \Omega_i) \frac{(kR)^3}{1-k^3}$$

Torque inner wall: - Torque outer wall:

$$= 8\pi\mu (\Omega_i - \Omega_o) \frac{(kR)^3}{1-k^3}$$

VELOCITY DISTRIBUTION:

$$v_\theta = \frac{kR}{1-k^3} \left[(\Omega_o - \Omega_i k^3) \left(\frac{r}{kR} \right) - (\Omega_o - \Omega_i) \left(\frac{kR}{r} \right)^3 \right] \sin \theta$$

3B.4 Creeping flow between two concentric spheres (Fig. 3B.4). A very viscous Newtonian fluid flows in the space between two concentric spheres, as shown in the figure. It is desired to find the rate of flow in the system as a function of the imposed pressure difference. Neglect end effects and postulate that v_θ depends only on r and θ with the other velocity components zero.

(a) Using the equation of continuity, show that $v_\theta \sin \theta = u(r)$, where $u(r)$ is a function of r to be determined.

(b) Write the θ -component of the equation of motion for this system, assuming the flow to be slow enough that the $[\mathbf{v} \cdot \nabla \mathbf{v}]$ term is negligible. Show that this gives

$$0 = -\frac{1}{r} \frac{\partial \mathcal{P}}{\partial \theta} + \mu \left[\frac{1}{\sin \theta} \frac{1}{r^2} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) \right] \quad (3B.4-1)$$

(c) Separate this into two equations

$$\sin \theta \frac{d\mathcal{P}}{d\theta} = B; \quad \frac{\mu}{r} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) = B \quad (3B.4-2, 3)$$

where B is the separation constant, and solve the two equations to get

$$B = \frac{\mathcal{P}_2 - \mathcal{P}_1}{2 \ln \cot \frac{1}{2} \epsilon} \quad (3B.4-4)$$

$$u(r) = \frac{(\mathcal{P}_1 - \mathcal{P}_2)R}{4\mu \ln \cot(\epsilon/2)} \left[\left(1 - \frac{r}{R} \right) + \kappa \left(1 - \frac{R}{r} \right) \right] \quad (3B.4-5)$$

where $\mathcal{P}_1$ and $\mathcal{P}_2$ are the values of the modified pressure at $\theta = \epsilon$ and $\theta = \pi - \epsilon$, respectively.

(d) Use the results above to get the mass rate of flow

$$w = \frac{\pi(\mathcal{P}_1 - \mathcal{P}_2)R^3(1 - \kappa)^3 \rho}{12\mu \ln \cot(\epsilon/2)} \quad (3B.4-6)$$

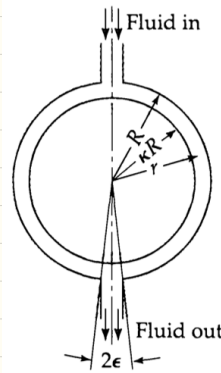


Fig. 3B.4. Creeping flow in the region between two stationary concentric spheres.

postulates:

- S.S
- constant μ, ρ
- neglect end effects

Velocity: $\langle \mathbf{v} \rangle = v_r = v_\phi = 0$

$$v_\theta = (r, \theta, \phi, t)$$

$$v_\theta = v_\theta(r, \theta)$$

a) Continuity equation:

$$\frac{\partial}{\partial \theta} (\rho v_\theta \sin \theta) = 0$$

where $\rho v_\theta \sin \theta$ not a function of θ

$$\Rightarrow \boxed{v_\theta \sin \theta = u(r)}$$

b) Equation of motion:

$$r\text{-comp: } -\frac{\rho v_\theta^2}{r} = -\frac{\partial \mathcal{P}}{\partial r} - \frac{2\mu}{r^2} \frac{\partial v_\theta}{\partial \theta} + \rho g_r$$

$$\theta\text{-comp: } \rho \frac{v_\theta}{r} \frac{\partial v_\theta}{\partial \theta} = -\frac{1}{r} \frac{\partial \mathcal{P}}{\partial \theta} + \mu \left[\frac{1}{r^2} \frac{\partial}{\partial r} \left(r^2 \frac{\partial v_\theta}{\partial r} \right) + \frac{1}{r^2 \sin \theta} \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial v_\theta}{\partial \theta} \right) - \frac{v_\theta}{r \sin^2 \theta} \right] + \rho g_\theta$$

$$\phi\text{-comp: } 0 = -\frac{1}{r \sin \theta} \frac{\partial \phi}{\partial \theta} + \mu \phi$$

$$\star \quad g_r = -g \cos \theta, \quad g_\theta = g \sin \theta, \quad g_\phi = 0$$

$$\text{in } 0 \leq \theta \leq \frac{\pi}{2} \rightarrow \cos \theta \Rightarrow (+), \quad \frac{\pi}{2} < \theta \leq \pi \rightarrow \cos \theta \Rightarrow (-)$$

$$\Rightarrow g_\theta = 0$$

$$\theta\text{-comp: } v_\theta = \frac{u(r)}{\sin \theta} \Rightarrow \frac{\partial v_\theta}{\partial \theta} = -\frac{\cos \theta}{\sin^2 \theta} u(r)$$

$$\frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial v_\theta}{\partial \theta} \right) = \frac{\partial}{\partial \theta} \left(-\frac{\cos \theta}{\sin \theta} u(r) \right) = u(r) \left(\frac{\sin^2 \theta + \cos^2 \theta}{\sin^2 \theta} \right) = \frac{u(r)}{\sin^2 \theta}$$

$$\frac{1}{r^2 \sin \theta} \cdot \frac{\partial}{\partial \theta} \left(\sin \theta \frac{\partial v_\theta}{\partial \theta} \right) = \frac{1}{r^2 \sin \theta} \cdot \frac{u(r)}{\sin^2 \theta} = \frac{1}{r^2 \sin^3 \theta} \cdot \frac{u(r)}{\sin \theta} = \frac{v_\theta}{r^2 \sin^2 \theta}$$

$$\Rightarrow -\mu \frac{\cos \theta}{\sin \theta} u^2(r) = -\frac{1}{r} \frac{\partial p}{\partial \theta} + \mu \left[\frac{1}{r^2 \sin \theta} \cdot \frac{\partial}{\partial r} \left(r^2 \frac{\partial u}{\partial r} \right) + \cancel{\frac{v_\theta}{r^2 \sin^3 \theta}} - \cancel{\frac{v_\theta}{r^2 \sin^3 \theta}} \right] + \mu g_\theta$$

$u(r)$ is small in creeping flow $\Rightarrow u^2(r)$ very small $\Rightarrow$ assume $u^2(r) \approx 0$

$$\star \text{ let } \beta = p - p_0$$

$$\Rightarrow 0 = -\frac{1}{r} \frac{\partial \beta}{\partial \theta} + \mu \left[\frac{1}{r^2 \sin \theta} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) \right]$$

c) Find $u(r)$:

$$\frac{1}{r} \frac{\partial \beta}{\partial \theta} = \mu \left[\frac{1}{r^2 \sin \theta} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) \right]$$

$$\text{Separate: } \underbrace{\sin \theta \frac{d\beta}{d\theta}}_{f(\theta)} = \underbrace{\frac{\mu}{r} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right)}_{h(r)} = \beta \text{ const}$$

$$\text{same for } \beta: \sin \theta \frac{d\beta}{d\theta} = \beta \Rightarrow \int d\beta = \int \frac{\beta}{\sin \theta} d\theta$$

$$\Rightarrow \Delta \beta = (\beta_1 - \beta_2) = -\beta \int_{\frac{\pi}{2}}^{\pi-\frac{\pi}{2}} \frac{1}{\sin \theta} d\theta = -\beta \ln \left[\tan \left(\frac{\theta}{2} \right) \right]_{\frac{\pi}{2}}^{\pi-\frac{\pi}{2}}$$

$$\Rightarrow -\beta \ln \left[\frac{\tan \left(\frac{\pi-\frac{\pi}{2}}{2} \right)}{\tan \left(\frac{\frac{\pi}{2}}{2} \right)} \right] = -\beta \ln \left(\frac{\cot \left(\frac{\pi}{2} \right)}{\tan \left(\frac{\pi}{2} \right)} \right)$$

$$\Rightarrow -\beta \ln \left[\frac{\cos(\frac{\epsilon}{2}) / \sin(\frac{\epsilon}{2})}{\sin(\frac{\epsilon}{2}) / \cos(\frac{\epsilon}{2})} \right]$$

$$\Rightarrow -\beta \ln \left[\frac{\cos^2(\frac{\epsilon}{2})}{\sin^2(\frac{\epsilon}{2})} \right] = -\beta \ln [\cot^2(\frac{\epsilon}{2})]$$

$$\Rightarrow \Delta \mathcal{F} = -2\beta \ln [\cot(\frac{\epsilon}{2})]$$

$$\text{for } u(r), \text{ let } E(\epsilon) = 2 \ln [\cot(\frac{\epsilon}{2})] = -\ln(\tan^2(\frac{\epsilon}{2}))$$

$$\Rightarrow \beta = \frac{-\Delta \mathcal{F}}{2 \ln(\cot(\frac{\epsilon}{2}))} = \frac{-\Delta \mathcal{F}}{E(\epsilon)}$$

$$\text{and } \frac{\mu}{r} \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) = \beta = -\frac{\Delta \mathcal{F}}{E(\epsilon)} \Rightarrow \frac{d}{dr} \left(r^2 \frac{du}{dr} \right) = -\frac{\Delta \mathcal{F}}{\mu E(\epsilon)} r$$

$$\Rightarrow r^2 \frac{du}{dr} = -\frac{\Delta \mathcal{F}}{2\mu E(\epsilon)} r^2 + C_1 \Rightarrow \frac{du}{dr} = -\frac{\Delta \mathcal{F}}{2\mu E(\epsilon)} + \frac{C_1}{r^2}$$

$$u(r) = -\frac{\Delta \mathcal{F}}{2\mu E(\epsilon)} \cdot r - \frac{C_1}{r} + C_2$$

$$\text{B.C.'s} \quad \begin{array}{lll} r=KR & V_A=0 & u(r)=0 \\ r=R & V_B=0 & u(r)=0 \end{array}$$

$$\Rightarrow 0 = -\frac{\Delta \mathcal{F}}{2\mu E(\epsilon)} \cdot KR - \frac{C_1}{KR} + C_2$$

$$0 = -\frac{\Delta \mathcal{F}}{2\mu E(\epsilon)} \cdot R - \frac{C_1}{R} + C_2$$

$$\text{Subtract: } \frac{\Delta \mathcal{F}}{2\mu E(\epsilon)} R(1-K) + \frac{1}{K} \left(1 - \frac{1}{K}\right) C_1 = 0$$

$$\Rightarrow C_1 = \frac{\Delta \mathcal{F}}{2\mu E(\epsilon)} \cdot R^2 K \quad C_2 = \frac{\Delta \mathcal{F}}{2\mu E(\epsilon)} \cdot R(K+1)$$

$$\Rightarrow u(r) = -\frac{\Delta \mathcal{F}}{2\mu E(\epsilon)} r - \frac{\Delta \mathcal{F}}{2\mu E(\epsilon)} \cdot \frac{R^2 K}{r} + \frac{\Delta \mathcal{F}}{2\mu E(\epsilon)} \cdot R(K+1)$$

$$\Rightarrow u(r) = \frac{R \Delta \mathcal{F}}{2\mu E(\epsilon)} \left[\left(1 - \frac{r}{R}\right) + K \left(1 - \frac{R}{r}\right) \right], \text{ where } E(\epsilon) = 2 \ln [\cot(\frac{\epsilon}{2})] = \ln(\cot^2(\frac{\epsilon}{2})) = -\ln(\tan^2(\frac{\epsilon}{2}))$$

d) Mass flow rate: $\dot{m} = \rho \dot{Q}$

axisymmetric flow rate:

$$\begin{aligned}\dot{Q} &= \int v_r dA = \int_0^{2\pi} \int_{kr}^R v \sin\theta r dr d\theta \\ &= \int_0^{2\pi} \int_{kr}^R u(r) r dr d\theta = 2\pi \int_{kr}^R u(r) r dr \\ &= \frac{\pi R \Delta p}{4 \mu E(\epsilon)} \int_{kr}^R \left[\left(r - \frac{r^2}{R} \right) + k(r-R) \right] dr\end{aligned}$$

integrate:

$$\begin{aligned}\dot{Q} &= \frac{\pi R \Delta p}{4 \mu E(\epsilon)} \left[\frac{r^2}{2} - \frac{r^3}{3R} + k \left(\frac{r^2}{2} - rR \right) \right]_{kr}^R \\ &= \frac{\pi R \Delta p}{4 \mu E(\epsilon)} \left[\frac{R^2}{2} (1-k^2) - \frac{R^2}{3} (1-k^3) + \frac{R^2}{2} k(1-k^2) - R^2 k(1-k) \right] \\ &= \frac{\pi R^3 \Delta p}{6 \mu E(\epsilon)} (1-3k+3k^2-k^3) = \frac{\pi R^3 \Delta p}{6 \mu E(\epsilon)} (1-k)^3\end{aligned}$$

mass flow rate: $\dot{m} = \frac{\pi R^3 \Delta p}{6 \mu E(\epsilon)} (1-k)^3 = \frac{\pi (p_1 - p_2) R^3 (1-k)^3}{12 \mu L \cot(\frac{\epsilon}{2})}$